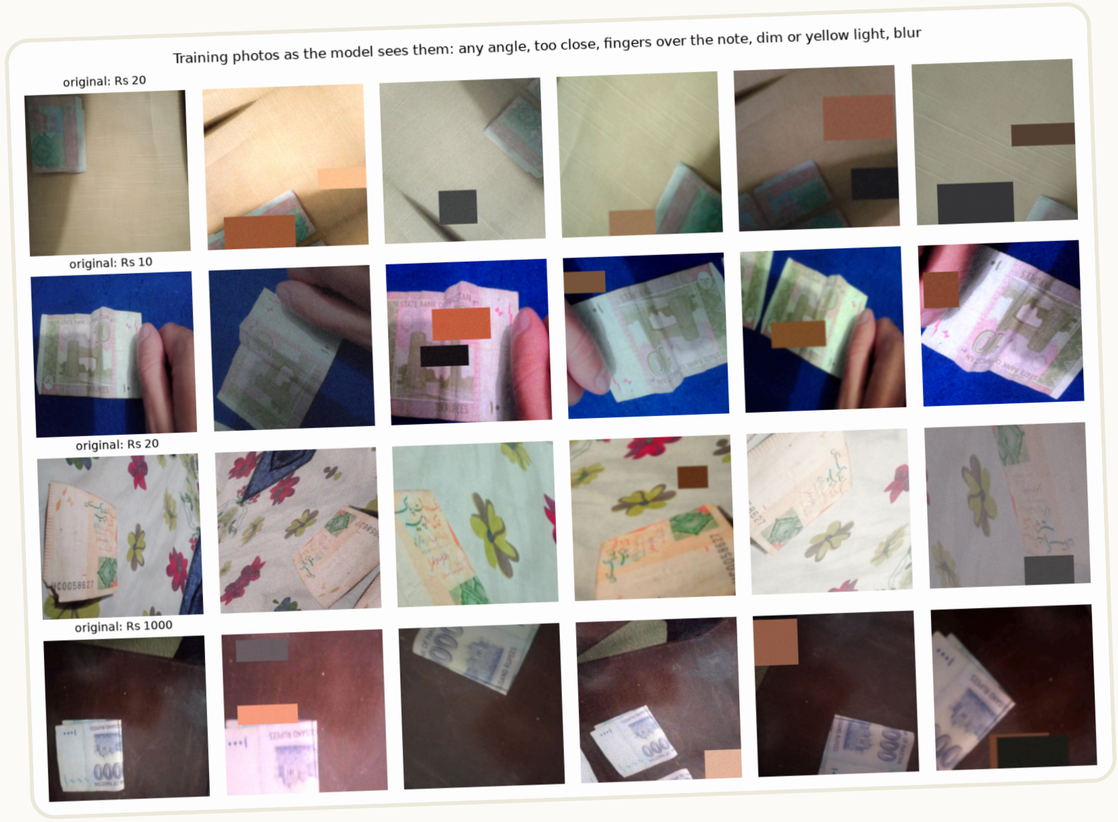
DEEP LEARNING · SEMESTER PROJECT

Pakistani Banknote Reader

Point your phone at a banknote. It says the value out loud, in Urdu.

Hassan Ali Mehdi · BSCS-7A · Superior University, Lahore
Instructor: Dr. Munir Ahmad



A blind person pays in cash every day. Every note is a question.

1.1M

blind people in Pakistan (GBD 2017)

7

notes in daily use, several close in size and colour

Rs 4,500

lost in one 500 / 5,000 mix-up

Today the answer is: ask a stranger, and trust them.

A web page, not an app

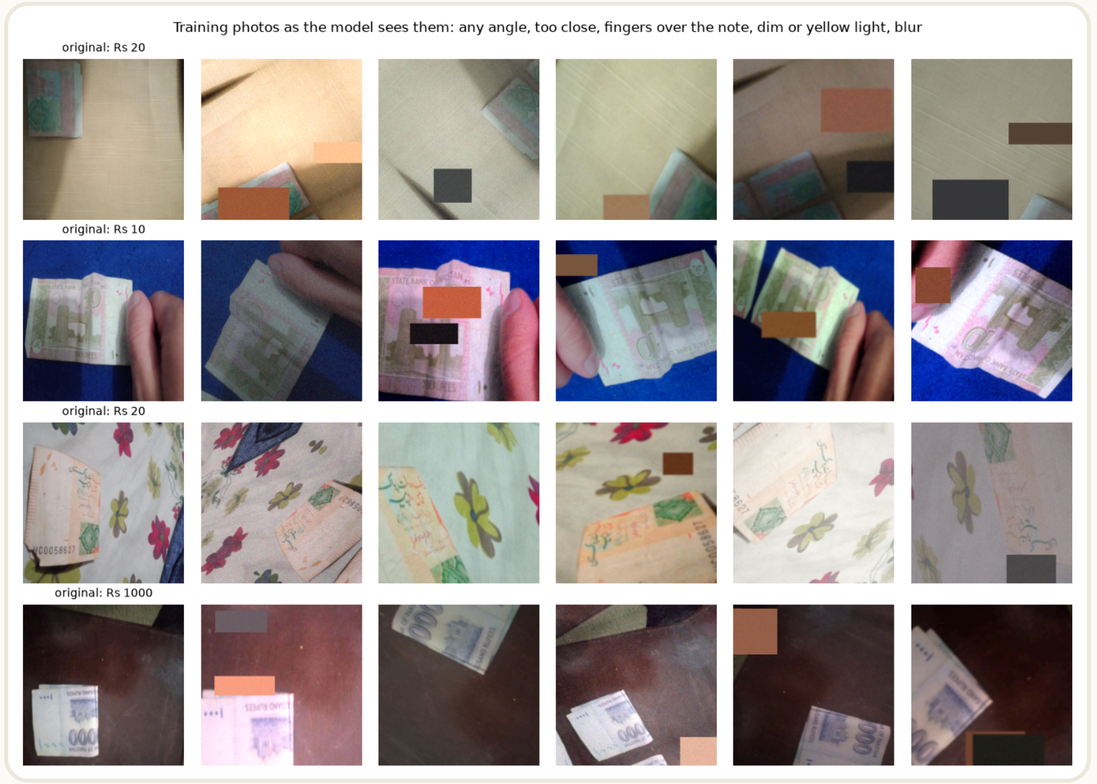
- Reads the live camera **4 times a second**
- Speaks the value in **Urdu** or English
- Stays **silent when unsure**: a wrong amount is worse than none
- Works **offline** once loaded, any phone browser
- Vibration code and screen-reader support



"Five hundred rupees"

Data: 3,611 real phone photos, 15 classes

- Kaggle **Pakistan Image Currency Dataset** (CC0)
- 7 notes × **front and back** = 14 classes
- + **"no note"**: 280 photos + 1,000 everyday Imagenette photos
- Folded, in hand, half visible, on bedsheets
- Split 70 / 15 / 15, near-copies kept together



Before trusting any number: are the test photos really new?

339 / 402

test photos in a second public dataset are near-copies of the first

- Perceptual hash of every photo (64 bits)
- Photos within 8 bits = the same photo
- Only **63** genuinely new photos left
- Without this check: near-perfect, and meaningless

Training for how blind people actually hold notes



Three models: accuracy vs speed

MODEL	PARAMS	SAME DATASET	OTHER PEOPLE	CPU
CNN from scratch	2.36M	99.6%	82.5%	88 ms
MobileNetV3-Large	4.22M	100.0%	92.1%	15 ms
MobileNetV4-Conv-Medium	8.45M	99.9%	92.1%	37 ms

Same data, same recipe. Highlighted: deployed (best on validation).

The honest numbers

100%

unseen photos from the same dataset

92%

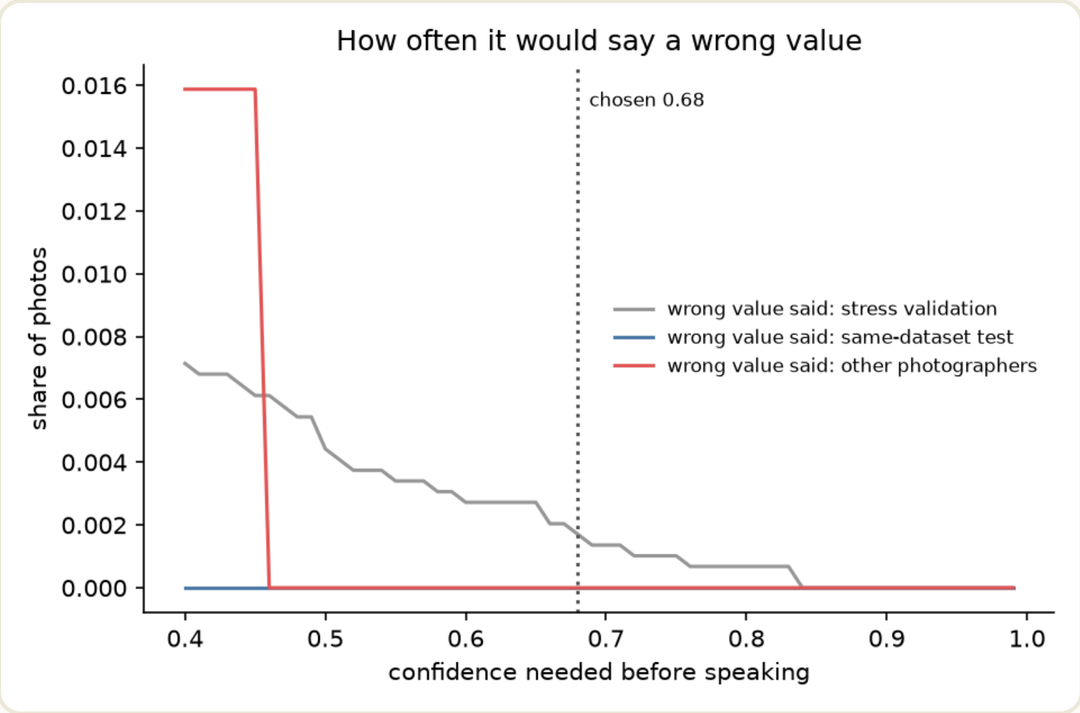
photos by other photographers (63)

0%

wrong amounts spoken on the photos by other people

The same-dataset score mostly shows how alike those photos are.
The other two are what a user would get.

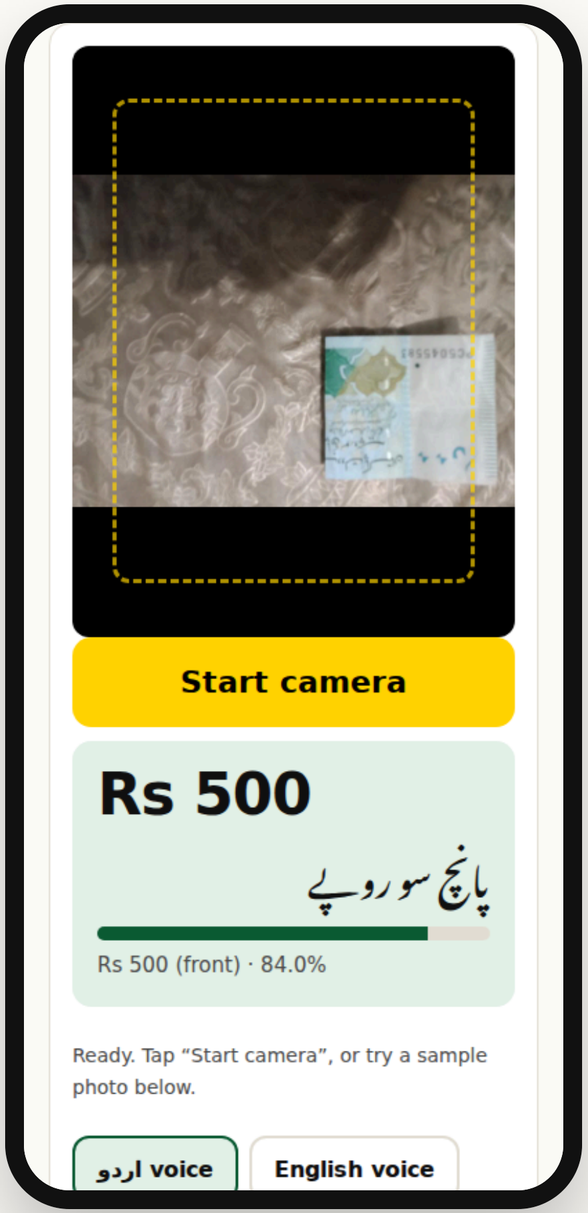
When to speak, and when to stay silent



- Validation photos look too easy: any threshold seems safe
- So: calibrate on a **stress test**, validation photos × 4 augmentations
- Threshold **0.68**: at most 1 wrong value in 500 there
- Plus: same answer in **3 frames in a row**

Live demo

- Open **https://boltanote.vercel.app** on any phone
- Tap **Start camera**
- Hold a note ~20 cm away
- It speaks: پانچ سو روپے



Limits, and what's next

Limits

- A few physical notes per value in training
- 2005–2008 designs; the new series needs new photos
- One note at a time; not a counterfeit detector

Next

- Worn, torn and taped notes
- Count a whole handful of notes
- Test with blind users

Everything is open

- Try it: **https://boltanote.vercel.app**
- Code, weights, splits: **github.com/Hassan-Ali-Mehdi-3024/boltanote**
- Report, slides, model card: linked on the site

Thank you, Dr. Munir Ahmad.

شکریہ